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Research Article

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A Compact Microstrip Patch Antenna for X-Band Applications

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Abstract*:* Nowadays, microstrip antennas are widely used in wireless applications. We have so many bands of frequency and we can list out L, S, X, Ka etc. For this work, we will be focusing on X bands only. The range of frequency of X bands is from 8 to 12 GHz. Design and simulate a compact microstrip patch antenna for X band applications are our main goals for this topic so that we will find out good results of antenna. The software used in this paper is HFSS 15 (High Frequency Structure Simulator 15). The proposed antenna is a rectangular microstrip patch with the dimensions of 6mm x 5mm using as substrate material FR4, dielectric constant 4.4 and operating frequency 11 GHz. After simulation, we got a gain equal to 2.8 dB, bandwidth equals to 700 MHz, directivity equals to 3.96 dB, VSWR (Voltage Standing Wave Ratio) equals to 1.24. These results show that antenna is very suitable for X bands applications (satellite communication, wireless computer networks, etc.)

Keywords: Patch antenna, X-bands, HFSS, Resonant frequency

Introduction

Today, with the explosion of telecommunications, some constraints are found in commercial applications wireless [1,2], patch antennas have been proposed to meet to these requirements, The past decade has witnessed the use rapidity of these antennas in communication and other systems satellite, radar. They combine small size, simplicity, ease of manufacturing and implementation, they adapt easily to flat and non-flat surfaces and exhibit a large sturdiness when mounted on rigid surfaces.

A printed antenna consists of a plate metal of any shape, called the radiating element, located on the upper face of a dielectric substrate. In another aspect, antennas are integral components in a communication system, communication which requires special study. While seeking to improve the performance of an antenna, it must be adapted to the most recent applications. The antenna should also meet the constraints of multiplying frequency bands, broadband and of integration. Finally, the characteristics of the antennas must be little influenced by the environment. Our study is more focus on X bands of frequency, specially at 11 GHz.

The patch antenna feeding technique is one of the most important parts in the design process, several techniques were then considered for this purpose [3], we can classify these techniques into two categories such as: Feeding by contact (Powered by a micro line, direct coaxial feed) and feeding by proximity (Powered by electromagnetic coupling, by opening coupling (slot) in the ground plane).

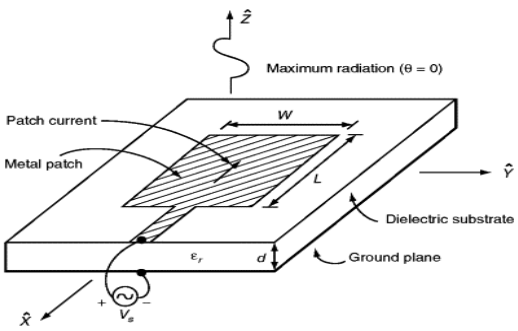

Figure 1: *Rectangular Microstrip Patch Antenna [4]*

The antenna is made by engraving of a printed circuit. Due to their manufacturing technology, these can be integrated into the electronic circuits by occupying a reduced volume and conforming to different types of surfaces. Their main advantage is their low manufacturing cost. Patch antennas are used in many applications from the X bands [4,5,6].

Design Process

To design any antenna, we need to consider some parameters like frequency, substrate material and thickness of the antenna. For our paper, we have designed at 11 GHz as operating frequency, FR4 material, 1.6 mm of thickness (h) and dielectric constant (ε_r) 4.4. Based on those parameters, we were able to calculate the dimensions of antenna with (W= width and L= Length) the equations [7, 11] below where C = 3 x 10^8 m/s

$$W = \frac{c}{2f} * \sqrt{\frac{2}{\varepsilon_r+1}} \tag{1}$$

$$\Delta L = 0.412h \frac{(\varepsilon_r+0.3)(\frac{W}{h}+0.264)}{\varepsilon_r-0.258)(\frac{W}{h}+0.8} \tag{2}$$

$$L = \frac{c}{2f}\left(\frac{\varepsilon_r+1}{2} - \frac{\varepsilon_r-1}{2}\sqrt{\left[1+12\frac{h}{w}\right]}\right)^{-\frac{1}{2}} - 2\Delta L \tag{3}$$

And about the dimensions of substrate, the following equations will help us know how to calculate

$$L_g = 6h + L \tag{4}$$

$$W_g = 6h + W \tag{5}$$

To feed an antenna, there are several types of feeding and we have chosen the micro line feeding. So, the equations below will show us how to calculate its dimensions [10].

$$Lf = \frac{L}{2}\sqrt{\varepsilon_r} \tag{6}$$

$$Wf = \frac{W}{2} \tag{7}$$

So, with those equations, we got the table 1

Table 1: Antennas Parameters

Symbols	Details	Values
f	frequency	11 GHz
h	thickness	1.6 mm
Wp	Width of patch	6 mm
Lp	Length of patch	4.5 mm
Wg	Width of ground	9 mm
Lg	Length of ground	7 mm
Wf	Width of feeding	0.6 mm
Lf	Length of feeding	2.25 mm
a		0.5 mm
b		1.5 mm
c		0.3 mm
d		1 mm

We used as software to design and simulate our antenna, HFSS 15. The figure 2 shows the proposed antenna in HFSS which is a compact microstrip patch antenna.

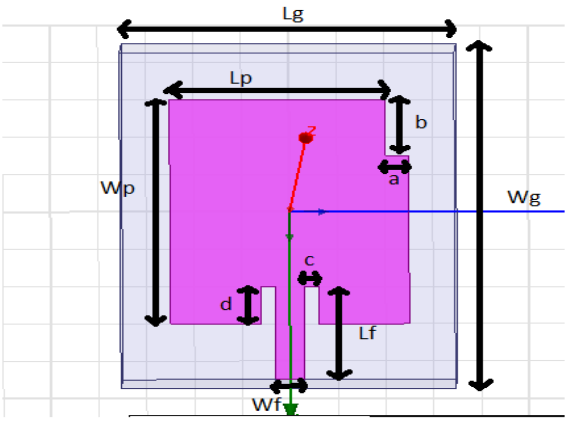

Figure 2: Compact microstrip patch antenna

The figure 2 shows us the compact microstrip patch antenna with a micro line as feeding. For a better result we had to cut (a x b) the patch to improve the value of parameters.

Results and Discussions

The simulation of the proposed antenna has been done through HFSS 15, and the results have been discussed in this part. Gradually, we found out the reflection coefficient, VSWR, gain, directivity and radiation pattern.

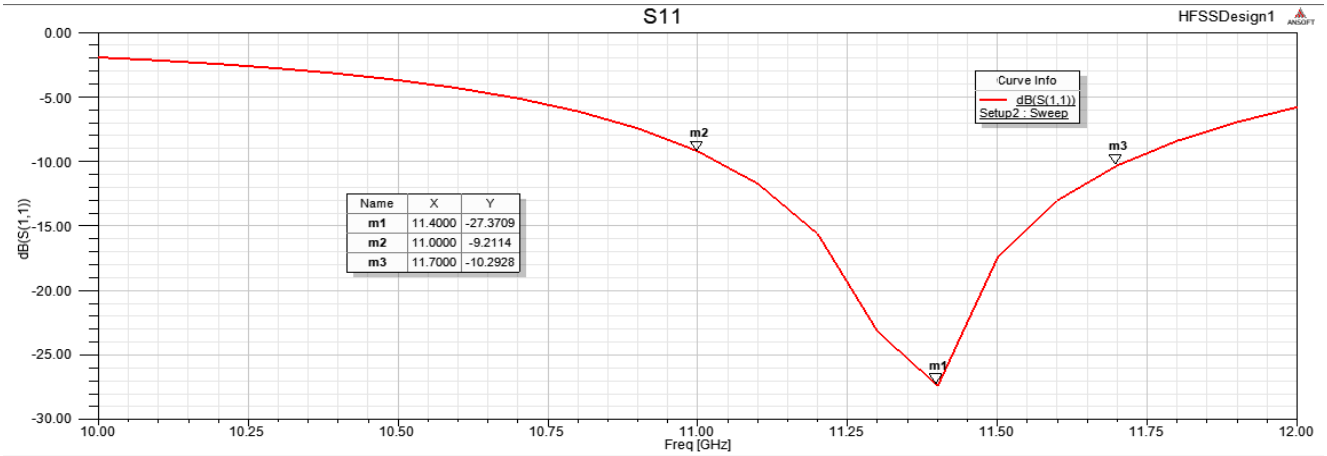

Figure 3: S11 of the compact microstrip patch antenna

The points m1, m2 and m3 on the graph represent respectively the place of the lower frequency, the operating frequency and the highest frequency. Based on that, we can calculate the range of frequencies where the antenna can operate which is call bandwidth (B). The equation [8] below will show us how:

$$B = F_h - F_l \tag{8}$$

Where F_h is the highest frequency and F_l is the lowest frequency

As we can in the figure 3, the bandwidth is equals to 700 MHz and the reflection coefficient is equals to -27.37 dB, that means only 0.18% of the transmitted power have been reflected back (to know the value of the transmitted power reflected back in watts (W) from the transmitted power reflected back in dB, we can use this formula below) [7,9]

$$P(W) = 10^{\frac{P(dB)}{10}} \tag{9}$$

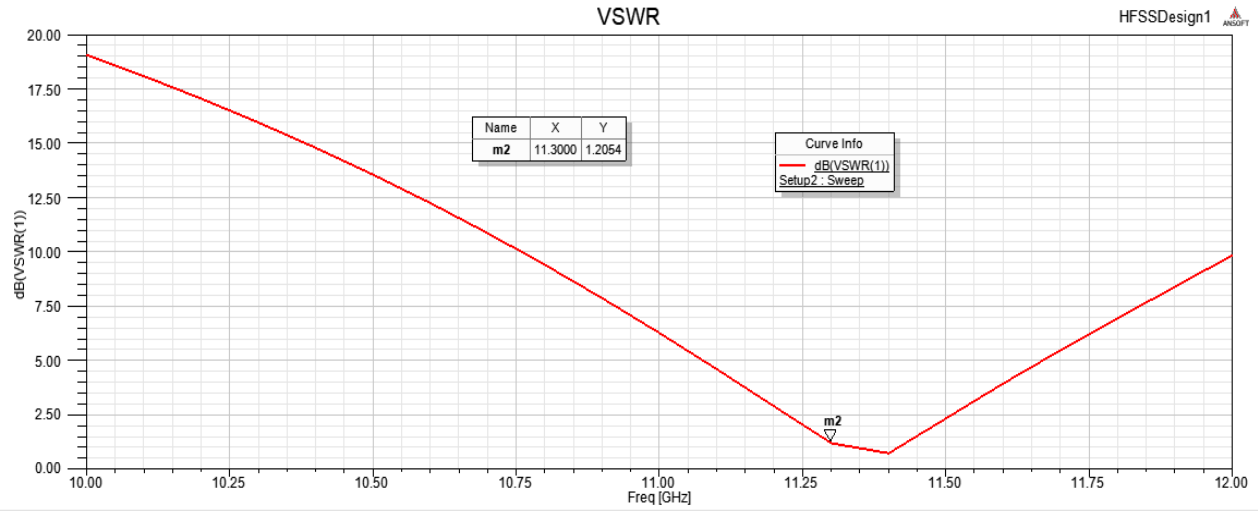

Figure 4: VSWR of the compact microstrip patch antenna

We can easily calculate VSWR from this following equation:

$$VSWR = \frac{|1+r|}{|1-r|} \tag{10}$$

Where, r is the reflection coefficient, based on that equation, we had VSWR equals to 1.24 and based on the figure 4, we had VSWR equals to 1.2064. we can the calculation and simulation result are almost same. VSWR helps us know the amount of the signal that has been reflected back due to mismatch components in the system.

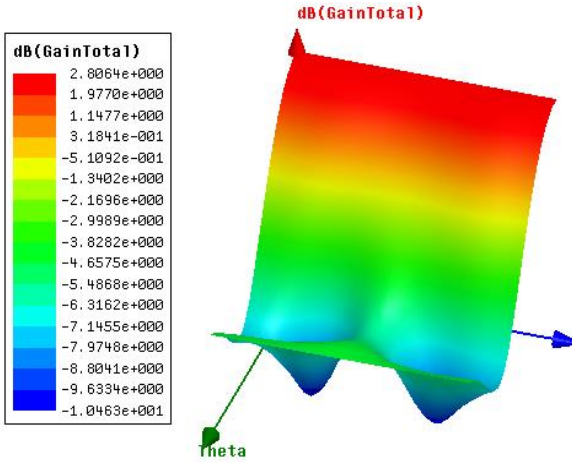

Figure 5: Gain of the compact microstrip patch antenna

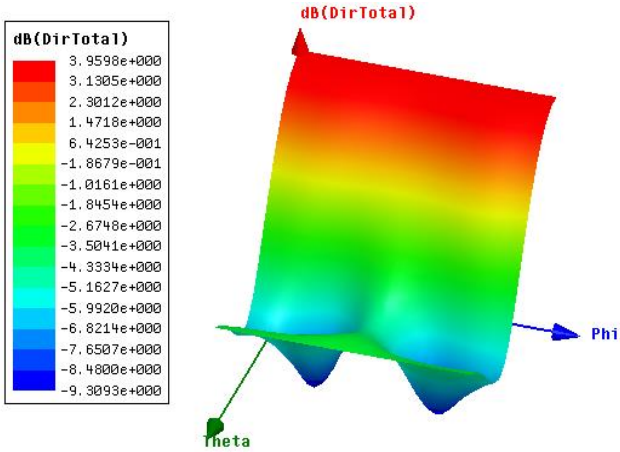

Figure 6: Directivity of the compact microstrip patch antenna

The figure 5 and figure 6 show respectively the highest value of the gain and directivity of the compact microstrip patch antenna which is 2.806 dB and 3.96 dB. Those values are good for an application that has a frequency equal to 11 GHz.

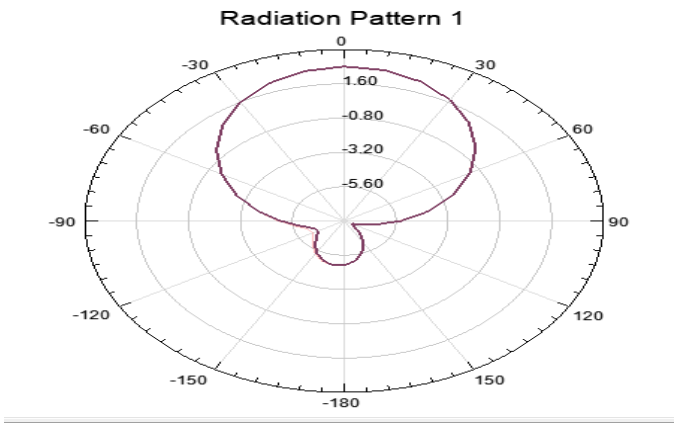

Figure 7: Radiation pattern of the compact microstrip patch antenna

The figure 7 is showing the radiation pattern of the proposed antenna and we can easily the quantity of the reflected power is very small. So, we only have the main lobe with a percentage of the transmitted power equals to 99.82 % and the back lobe with a percentage of the reflected power equals to 0.18%.

Table 2: Summary of results

Parameters	Values
Operating frequency	11.3 GHz
Dimensions of the patch	6mm x 5mm
Bandwidth	700 MHz
Gain	2.8 dB
Directivity	3.96 dB
VSWR	1.24
Reflection coefficient	-27.37

Conclusion

In this paper, authors have proposed a compact microstrip patch antenna for X band applications using HFSS 15 at 11 GHz of frequency. The proposed antenna can be suitable for wireless applications since it has a wideband of frequency. The table 2 shows the summary results of our proposed antenna. Our further work will consist to fabricate the antenna and compare the results with the simulated ones.

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Statements and Declarations

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Availability of Data	All data is only available in full in the results section of this paper
Software	HFSS 15 has been used in this paper and can be freely downloaded on: https://getintopc.com/softwares/simulators/ansys-hfss-15-0-3-x64-free-download-8524782/

Data Availability Statement

Openly Available Data	All data is provided in full in the results section of this paper
Citing Multiple Datasets	This paper is supported by multiple datasets which are openly available in the reference section

Sensitive Data	This paper has no restrictions and can be accessed without any condition
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